

Software Requirement Specification

Project : Fleet Management

Team 28 [Fleetix]

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Problem Statement

Open source route optimization using Google APIs and other optimization techniques for employee transportation, specifically addressing scenarios involving multiple pickups and drop-offs.

System Requirements

The core requirement involves efficiently transporting employees with multiple pickup and drop-off locations. This can be categorized into two primary scenarios:

- **Many-to-One** : Multiple employees are picked up from different locations and transported to a single destination (e.g., office or event).
- **Many-to-Many** : Special case of the first scenario, where employees are picked up from different locations and dropped off at various destinations, possibly across various times.

For implementing these, following are the smaller specific features, the client has requested :

- **Create a Road Network Graph** - Represent the road network for the city/town in a graph data structure. This graph will be used for route calculations.
- **Geocode Employee Locations** - Convert employee addresses to geographic coordinates (latitude and longitude) for mapping onto the road network.
- **Constraints** : Such as vehicle capacity, time windows, and maximum route duration should be considered in final routing.

Apart from this main problem, following are some suggested features (or additional considerations) from the client :

- **User Interface** : Develop a user-friendly interface for inputting employee data, visualizing routes, and managing transportation schedules.

The client has an ongoing project, which he would later integrate this code with. Thus, would prefer a tech stack similar to it. The following are the current product's tech stack :

- **Frontend** is in next (and typescript)
- **Backend** is in next.js
- **Database** is MongoDB

User Profiles

1. System Administrators

- Add / Remove / Edit drivers and employees
- Update constraints like vehicle capacity, routes, wait time and maximum acceptable delays

2. Drivers

- View routes (past and scheduled)
- Passenger details like pick-up location, drop-off location and contact details

3. Employees

- View routes (past and scheduled) along with co-passengers, driver and vehicle details
- Pickup/drop-off (location and time) preference settings

Use Cases

NOTE: Since the use case of the given problem statement is very specific with not many use cases for the users, we would consider any functionality doing an isolated task as a use case. While this deviates from the definition of “relation between actors”, it keeps up with the goal of breaking down the problem for better understanding and implementation.

S. No.	Use Case Name	Description	Release
1	Geocoding	Extracting (longitude, latitude) out of an address string using Google APIs	R1
2	Road Network	Develop / Extract (from a library) a detailed road network graph	R1
3	Employee Pick-up / Drop-off Location Preference	Option for the employees to set their preferred pick-up / drop-off location	R1
4	Routing (Many-to-One)	Choosing and implementing an optimal algorithm for routing (a singular vehicle) through the extracted longitudes and latitudes	R1
5	Visualise Route	Appropriately visualise the routes with detailing for driver and employees	R1
6	Adding Constraints	Fine tune the algorithm to work with predefined constraints like vehicle capacity, maximum delay and wait time	R2
7	Ride History and Reports	Employees and admins can view past rides, including routes taken and trip details.	R2
8	Many-to-Many Routing	Multiple vehicle routing	R2
9	User Interface	Develop interactive and user-friendly web-based applications for driver, employees and system administrator	R2

Use Case Description

UseCase Number	UC-001
Use Case Name	Geocoding
Overview	Extracting (longitude, latitude) out of an address string using Google APIs
Actors	System
Pre-Condition	Employee addresses are available for processing.
Flow	Main Flow 1. The system receives an address string from the input dataset. 2. The Google Geocoding API processes the address and returns corresponding latitude and longitude coordinates.
	Alternate Flow 2a. If the address could not be decoded by the Google API, appropriate errors are used. And system admins / employees would be prompted to reword the address string. *a. Any errors, at any stage, should prompt the admin / employee to reword the address string.
Post-Condition	Successfully geocoded locations are returned for route processing.

UseCase Number	UC-002
Use Case Name	Road Network
Overview	Develop / Extract (from a library) a detailed road network graph
Actors	System
Pre-Condition	Appropriate data library is active.
Flow	Main Flow 1. The system retrieves road network data. 2. The data is structured into appropriate structure and stored for future route calculations.
	Alternate Flow 1a. If the data fetch fails, the system retries fetching or uses a cached version (if available). After certain retries, the admin is notified.
Post-Condition	A structured road network graph is ready for routing operations.

UseCase Number	UC-003
Use Case Name	Employee Pick-up / Drop-off Location Preference
Overview	Option for the employees to set their preferred pick-up / drop-off location
Actors	Employee, System
Pre-Condition	The employee is registered in the system.
Flow	Main Flow 1. The employee logs into the system and navigates to the preferences section. 2. The employee inputs a preferred pickup and drop-off location. 3. The input is verified by the system against Google Geocoding API.
	Alternate Flow 3a. If the inputted address could not be geocoded, the employee would be prompted to enter a reworded address string; until a geocode is obtained. *a. Any errors, at any stage, should raise appropriate error messages and no backend change should go through.
Post-Condition	Employee preference(s) are saved for future routing.

UseCase Number	UC-004
Use Case Name	Routing (Many-to-One)
Overview	Choosing and implementing an optimal algorithm for routing (a singular vehicle) through the extracted longitudes and latitudes
Actors	System
Pre-Condition	Road network graphs and geocoded employee locations are available.
Flow	Main Flow 1. The system retrieves the locations of employees to be picked up. 2. The routing algorithm calculates the optimal path using Google Routes API.. 3. Route path is generated and stored.
	Alternate Flow NA
Post-Condition	The optimal Many-to-One route is generated and stored.

UseCase Number	UC-005
Use Case Name	Visualise Route
Overview	Appropriately visualise the routes with detailing for driver and employees
Actors	System, Driver, Employee
Pre-Condition	Routing is complete.
Flow	Main Flow 1. The system fetches the generated route. 2. The route is displayed on an interactive map interface. 3. Employees and drivers can view the planned route.
	Alternate Flow 2a. If the map service fails, textual directions are provided.
Post-Condition	Routes are successfully displayed for all relevant users.

UseCase Number	UC-006
Use Case Name	Adding Constraints
Overview	Fine tune the algorithm to work with predefined constraints like vehicle capacity, maximum delay and wait time
Actors	System, System Administrator
Pre-Condition	Routing system is operational.
Flow	Main Flow 1. The administrator defines constraints such as vehicle capacity and wait times. 2. The routing algorithm incorporates these constraints for future routing.
	Alternate Flow *a. If constraints make routing impossible, the admin is notified.
Post-Condition	Constraints are successfully applied to routing.

UseCase Number	UC-007
Use Case Name	Ride History and Reports
Overview	Employees and admins can view past rides, including routes taken and trip details.
Actors	Employee, System Administrator

Pre-Condition	The system has rides data stored
Flow	Main Flow 1. Users navigate to the ride history section. 2. The system fetches and displays past ride details. 3. Users can filter by date, location, or driver.
	Alternate Flow 1a. If no records exist, an appropriate message is displayed.
Post-Condition	Ride history is displayed successfully.

UseCase Number	UC-008
Use Case Name	Many-to-Many Routing
Overview	Multiple vehicle routing
Actors	System
Pre-Condition	Multiple vehicles and employee locations are available.
Flow	Main Flow 1. The system groups employees based on proximity and drop-off location. 2. Routes are assigned to multiple vehicles. 3. The system ensures optimal vehicle utilization.
	Alternate Flow *a. In case of any error, appropriate errors are displayed. The user may re-run the functionality.
Post-Condition	Many-to-Many routes are generated successfully.

UseCase Number	UC-009
Use Case Name	User Interface
Overview	Develop interactive and user-friendly web-based applications for driver, employees and system administrator
Actors	System, Users (Employees, Drivers, Admins)
Pre-Condition	Backend functionality is complete.
Flow	Main Flow 1. UI is designed and implemented for each user role. 2. Employees and drivers can access their respective routes, stops, passengers and vehicle details. 3. Admins can view the status of all vehicles, drivers, vehicles and routes.

	Alternate Flow NA
Post-Condition	Users can interact seamlessly with the system.

Use Case Diagram

